

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper

Total marks: 100

Section I – 75 marks (pages 2–28)

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–30
- Allow about 1 hour and 40 minutes for this part

Section II – 25 marks (pages 29–39)

- Attempt ONE question from Questions 31–35
- Allow about 45 minutes for this section

Section I

75 marks

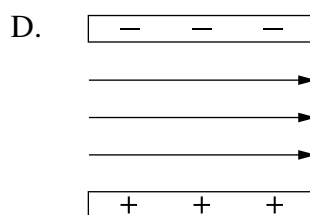
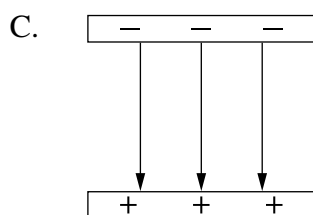
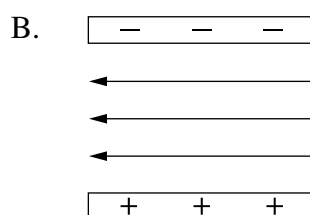
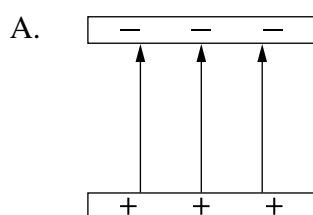
Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

- 1 Who was the main advocate for the distribution of DC electricity to cities?
- A. Edison
B. Faraday
C. Tesla
D. Westinghouse
- 2 Which of the following is an inertial frame of reference?
- A. A rocket during launch
B. A train travelling at a constant velocity
C. A car turning a corner at a constant speed
D. A lift slowing down as it approaches the ground floor
- 3 Which of the following correctly shows the electric field between two parallel, charged plates?



- 4 An astronaut with a mass of 75 kg lands on Planet *X* where her weight is 630 N.

What is the acceleration due to gravity (in m s^{-2}) on Planet *X*?

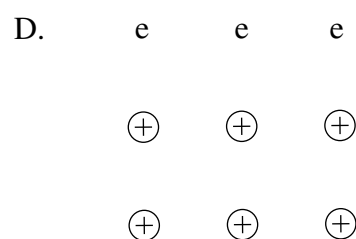
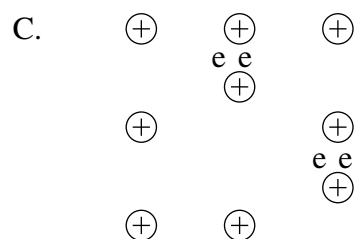
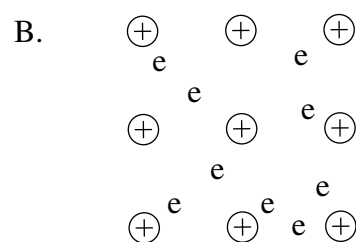
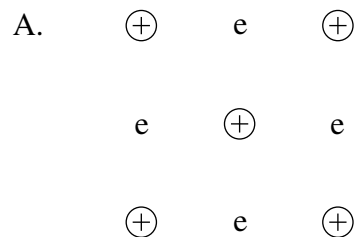
- A. 0.12
- B. 8.4
- C. 9.8
- D. 735

- 5 An electric current flows in undoped silicon.

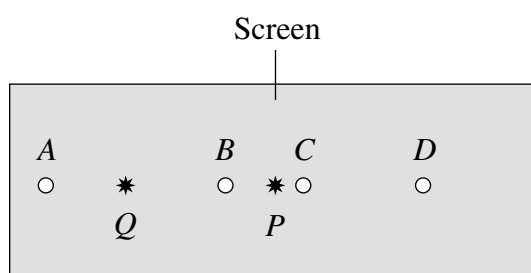
Which statement about this current is true?

- A. The number of electrons moving exceeds the number of holes moving.
- B. The number of holes moving exceeds the number of electrons moving.
- C. Equal numbers of holes and electrons are moving in the same direction.
- D. Equal numbers of holes and electrons are moving in opposite directions.

- 6 Which of the following best represents the structure of a metallic lattice?

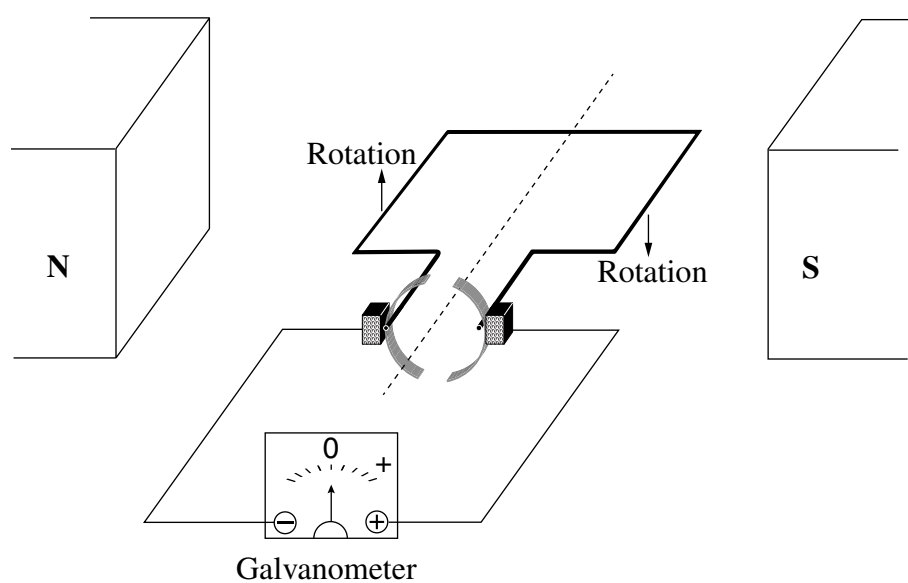


- 7 Which of the following is a correct statement about the Michelson–Morley experiment?
- A. It proved the existence of the luminiferous aether.
 - B. It compared the speeds of light rays travelling along parallel paths.
 - C. It was invalid because the equipment was not sufficiently sensitive.
 - D. It was unable to detect the motion of Earth through the luminiferous aether.
- 8 An electron is fired in a vacuum towards a screen. With no electric field being applied, the electron hits the screen at *P*. A uniform electric field is turned on and another electron is fired towards the screen from the same location, at the same velocity, striking the screen at point *Q*.



- With the electric field still turned on, a proton is fired towards the screen from the same starting point as the electrons and with the same velocity.
- At what point does the proton strike the screen?
- A. *A*
 - B. *B*
 - C. *C*
 - D. *D*
- 9 What does the Maltese cross apparatus demonstrate about cathode rays?
- A. They travel in straight lines.
 - B. They consist of a beam of electrons.
 - C. They are absorbed by a metal surface.
 - D. They are not affected by magnetic fields.

- 10 The diagram shows a model of a generator connected to a galvanometer.

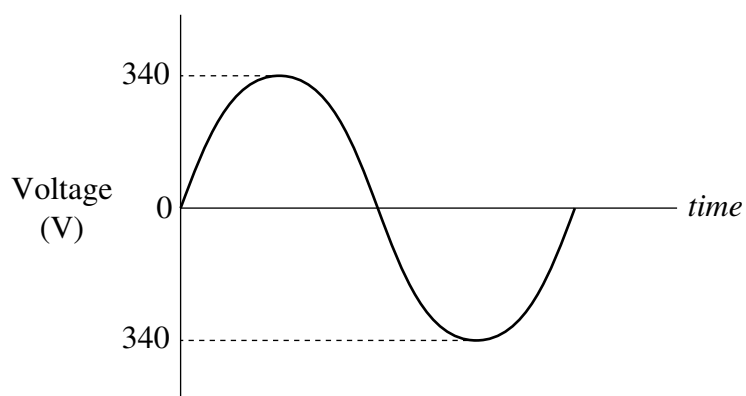
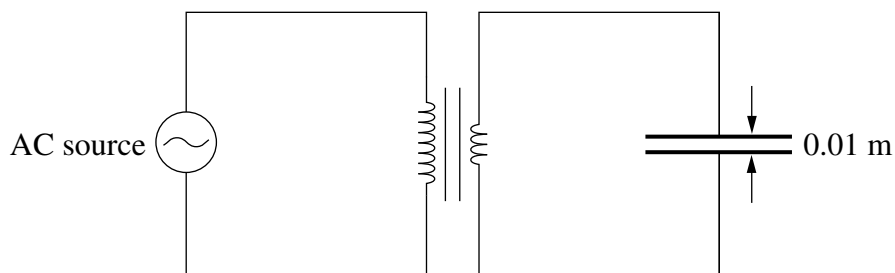


The loop is rotated continuously in a clockwise direction as viewed from the end nearest the galvanometer.

Which row of the table correctly identifies the type of generator and the movement of the needle of the galvanometer?

	<i>Type of generator</i>	<i>Movement of the needle</i>
A.	DC	Swings between 0 and +
B.	AC	Swings between – and 0
C.	DC	Swings between + and –
D.	AC	Swings between – and +

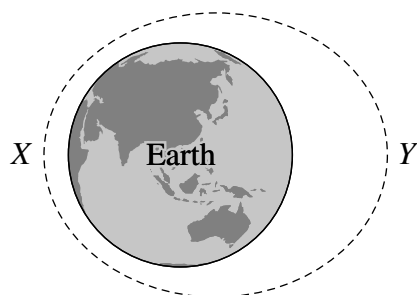
- 11 An AC source is connected to a transformer having a primary winding of 900 turns. Connected to the secondary winding of 450 turns is a pair of parallel plates 0.01 m apart.



What is the maximum field strength (in V m^{-1}) produced between the plates?

- A. 1.7
- B. 6.8
- C. 1.7×10^4
- D. 6.8×10^4

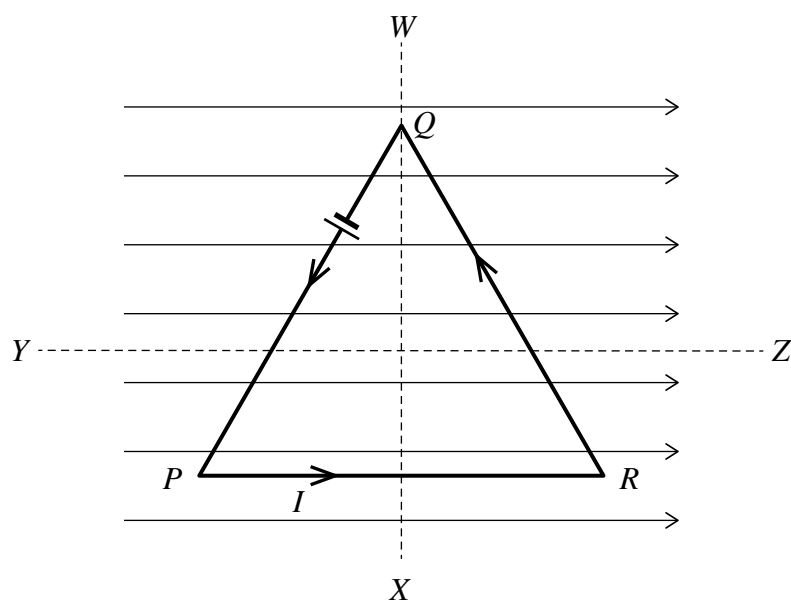
- 12 A satellite orbits Earth with an elliptical orbit that passes through positions X and Y .



Which row of the table correctly identifies the position at which the satellite has greater kinetic energy and the position at which it has greater potential energy?

	<i>Greater kinetic energy</i>	<i>Greater potential energy</i>
A.	X	X
B.	X	Y
C.	Y	X
D.	Y	Y

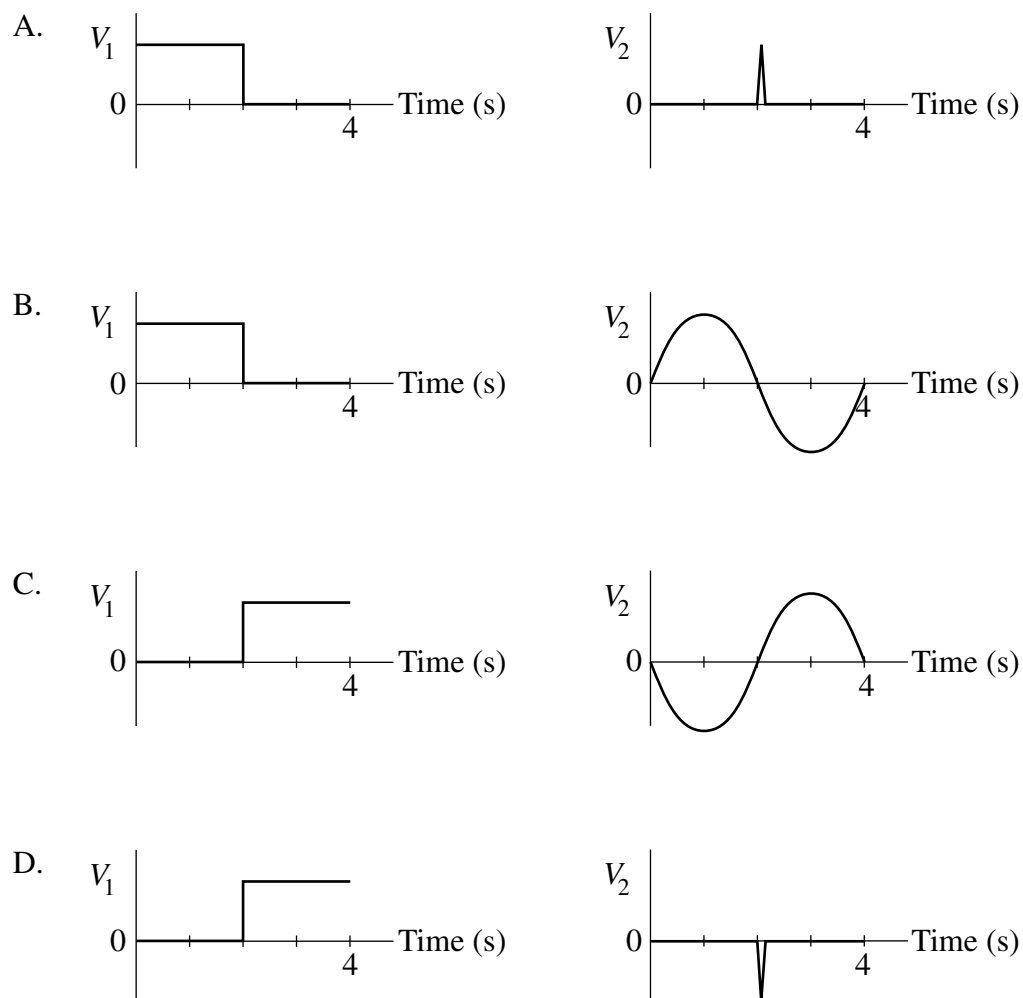
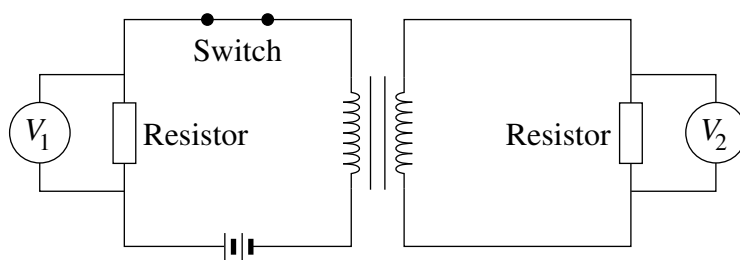
- 13 A triangular piece of wire is placed in a magnetic field as shown.



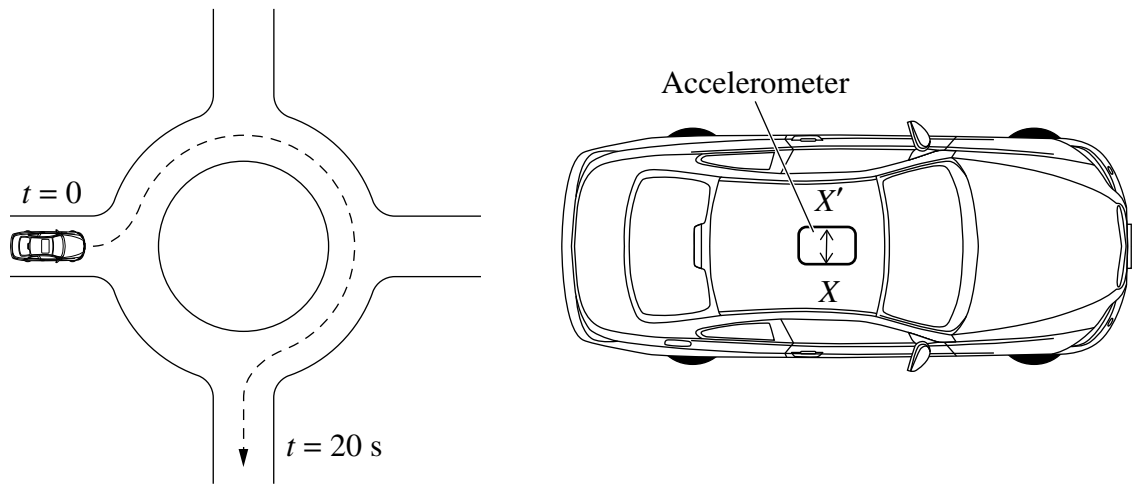
When current I is supplied as shown, how does the wire move?

	<i>Axis of rotation</i>	<i>Direction of movement</i>
A.	YZ	Q into page
B.	YZ	Q out of page
C.	WX	R into page
D.	WX	R out of page

- 14 The diagram shows a DC circuit containing a transformer.

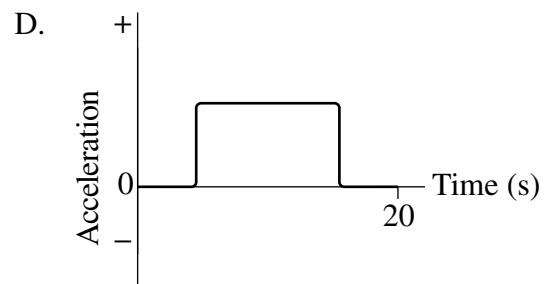
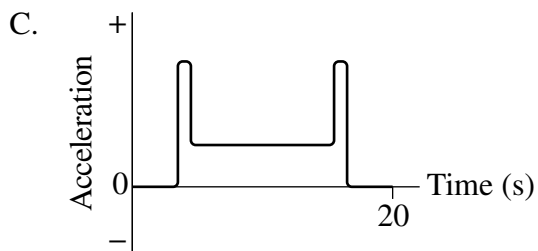
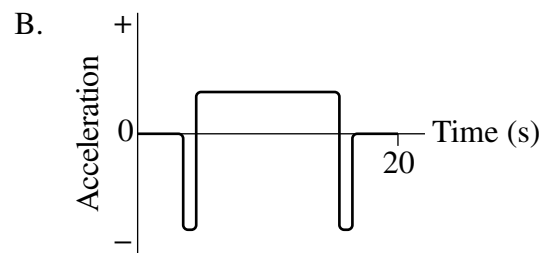
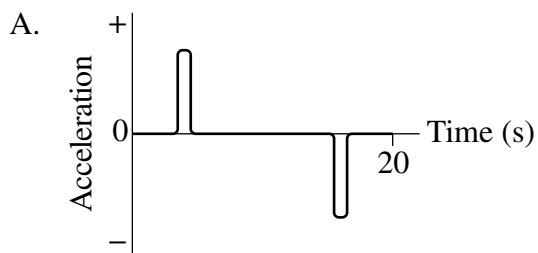


- 15 A car travelling at a constant speed follows the path shown.

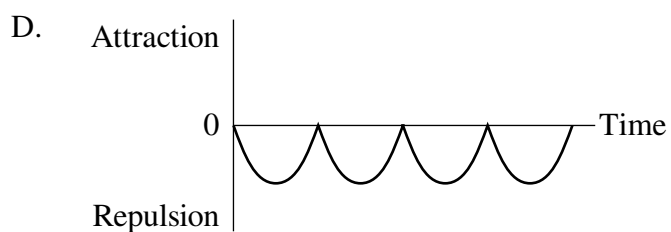
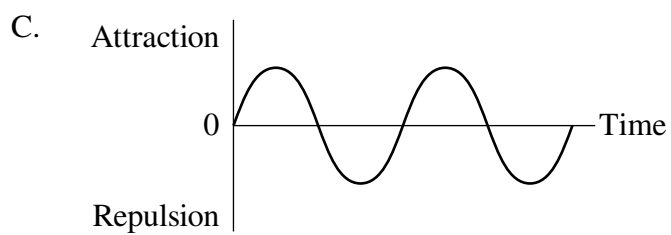
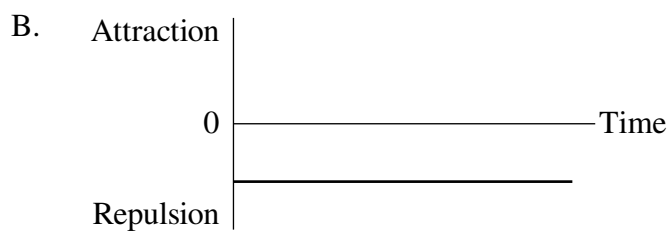
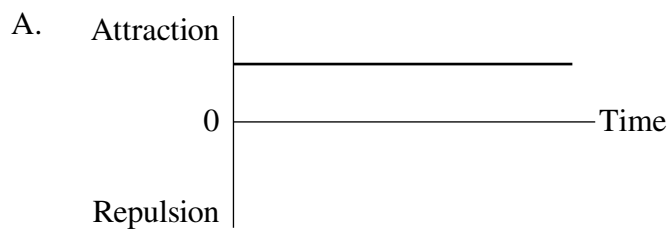


An accelerometer that measures acceleration along the $X-X'$ direction is fixed in the car.

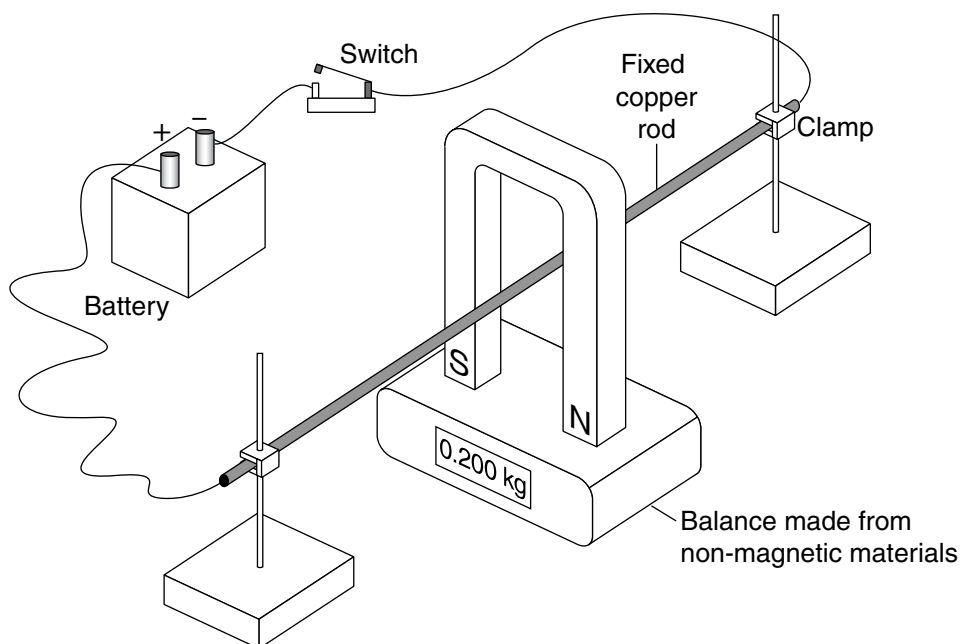
Which graph shows the measurements recorded by the accelerometer over the 20-second interval?



- 16 An AC supply is connected to a light bulb by two long parallel conductors as shown.



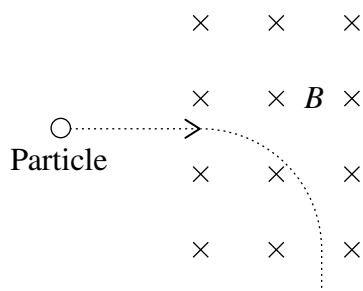
- 17 A magnet rests on an electronic balance. A rigid copper rod runs horizontally through the magnet, at right angles to the magnetic field. The rod is anchored so that it cannot move.



Which expression can be used to calculate the balance reading when the switch is closed?

- A. $0.200 \text{ kg} + BIl$
- B. $0.200 \text{ kg} + \frac{BIl}{9.8}$
- C. $0.200 \text{ kg} - BIl$
- D. $0.200 \text{ kg} - \frac{BIl}{9.8}$

- 18 A particle of mass m and charge q travelling at velocity v enters a magnetic field of magnitude B and follows the path shown.

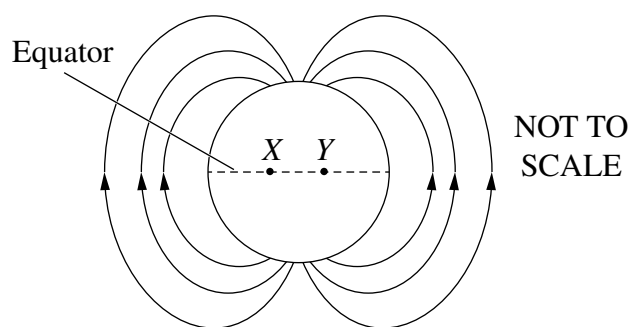


A second particle enters a magnetic field of magnitude $2B$ with a velocity of $\frac{1}{2}v$ and follows an identical path.

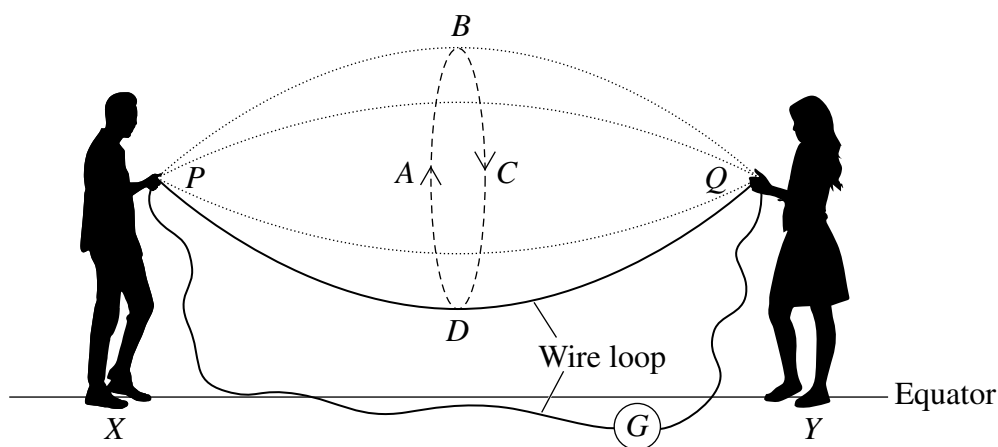
What is the mass and charge of the second particle?

	<i>Mass</i>	<i>Charge</i>
A.	m	q
B.	$\frac{1}{2}m$	$2q$
C.	$4m$	q
D.	m	$\frac{1}{2}q$

- 19 Earth's magnetic field is shown in the following diagram.



Two students standing a few metres apart on the equator at points X and Y , where Earth's magnetic field is parallel to the ground, hold a loop of copper wire between them. Part of the loop is rotated like a skipping rope as shown, while the other part remains motionless on the ground.



At what point during the rotation of the wire does the maximum current flow in a direction from P to Q through the moving part of the wire?

- A. A
- B. B
- C. C
- D. D

- 20** The length of a spaceship is measured by an observer to be 3.57 m as the spaceship passes with a velocity of $0.7c$.

At what velocity would the spaceship be moving relative to the observer if its measured length was 2.5 m?

- A. $0.490c$
- B. $0.707c$
- C. $0.714c$
- D. $0.866c$

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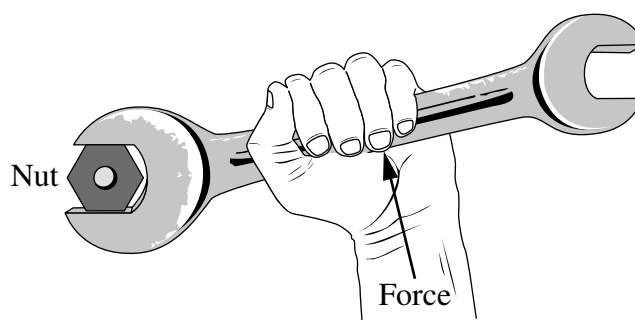
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Suggest TWO ways that the applied torque could be increased.

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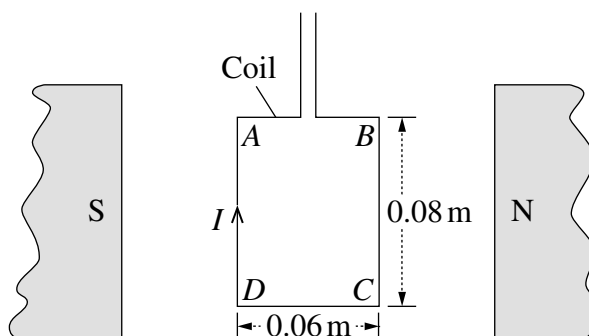
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- (b) A coil consisting of 15 turns is placed in a uniform 0.2 T magnetic field between two magnets. A current of 7.0 amperes flows in the direction shown.

3



Calculate the magnitude and direction of the torque produced by the side BC of the 15-turn coil.

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$$v = \sqrt{\frac{2GM}{r}}.$$

- (a) The radius of Mars is 3.39×10^6 m and its mass is 6.39×10^{23} kg. 2

Calculate the escape velocity from the surface of Mars.

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- (b) Using the law of conservation of energy, show that the escape velocity of an object is independent of its mass. 3

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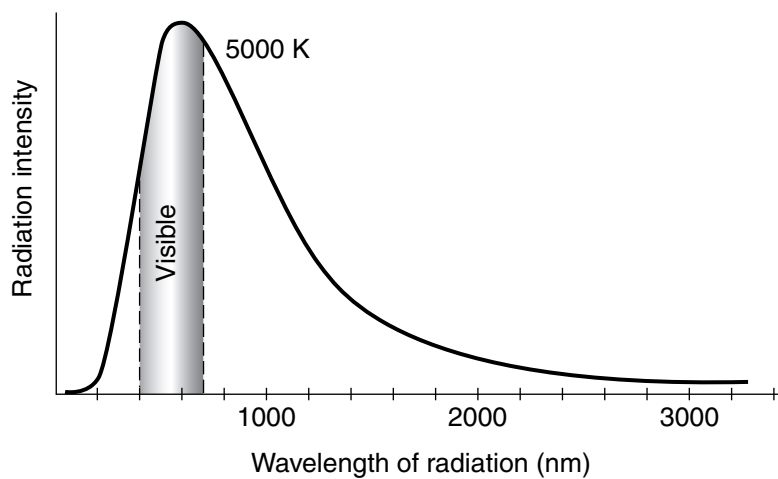
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Relate the information in the graph to the energy changes required for the shuttle to make a safe landing.

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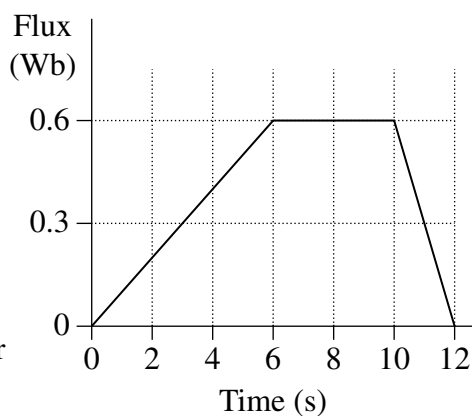
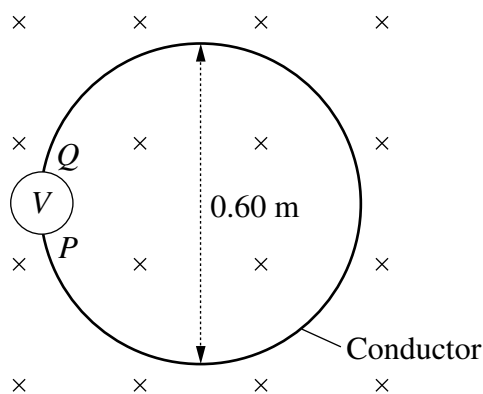
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- (a) Calculate the maximum magnetic field strength within the stationary loop during the 12-second interval. 2

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- (b) Calculate the maximum voltage generated in the circuit by the changing flux. In your answer, indicate the polarity of the terminals *P* and *Q* when this occurs. 3

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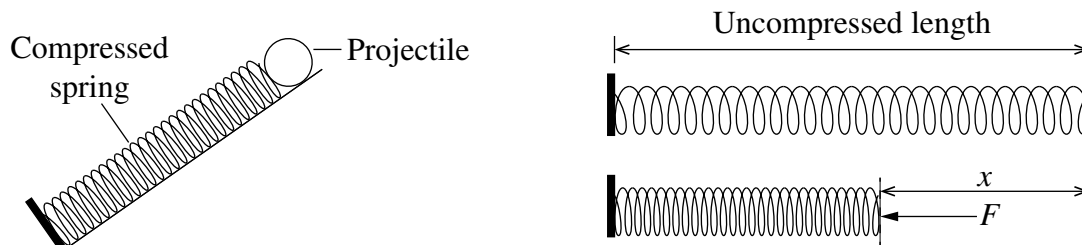
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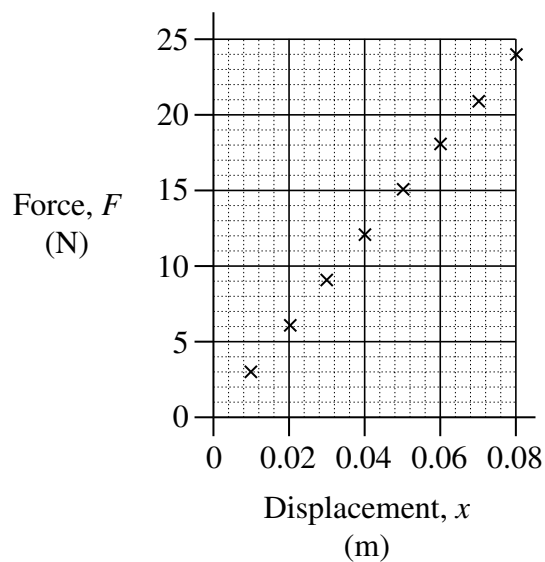
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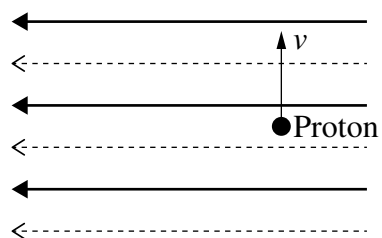


The potential energy stored in the compressed spring can be calculated from $E_p = \frac{1}{2}kx^2$, where k is the gradient of the force–displacement graph shown.



Question 29 continues on page 27

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KEY

Electric field

Magnetic field

Analyse the motion of the proton after the fields have been turned on.

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Physics

Section II

25 marks

Attempt ONE question from Questions 31–35

Allow about 45 minutes for this section

Answer parts (a)–(e) of one question in the Section II Writing Booklet. Extra writing booklets are available.

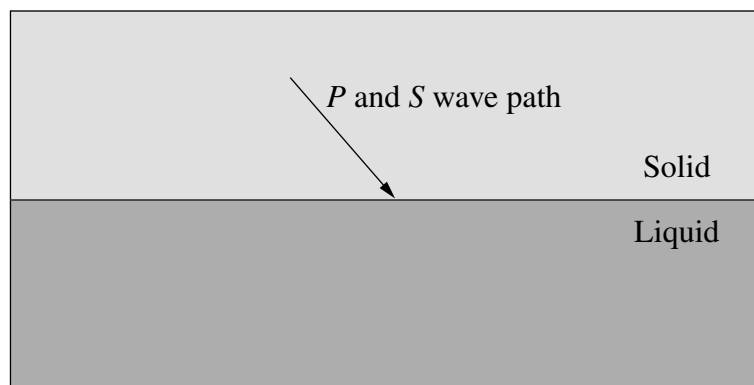
Show all relevant working in questions involving calculations.

	Pages
Question 31 Geophysics	30–31
Question 32 Medical Physics	32
Question 33 Astrophysics	33–34
Question 34 From Quanta to Quarks	35–36
Question 35 The Age of Silicon	37–39

Question 31 — Geophysics (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) Give TWO reasons why the scientific community was initially reluctant to accept the idea that Earth's plates are moving. **2**
- (ii) How do features associated with the magnetic properties of the oceanic crust support the theory of plate tectonics? **3**
- (b) (i) *P* waves and *S* waves are travelling through the same medium. Contrast the movement of particles associated with these waves. **2**
- (ii) Describe the behaviour of the *P* waves and *S* waves when they reach the boundary shown in the diagram. **3**



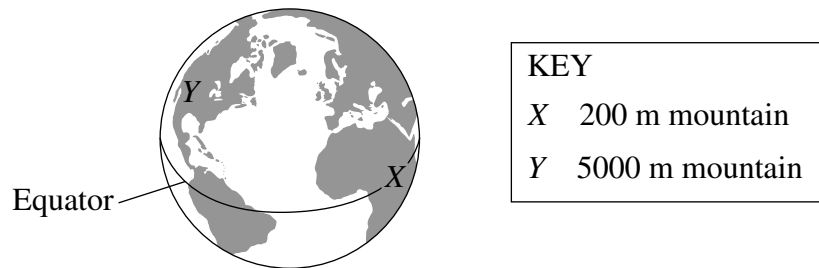
- (c) Describe TWO different methods that can be used to calculate the mass of Earth. In your answer, refer to the relevant equations. **4**

Question 31 continues on page 31

Question 31 (continued)

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) (i) Outline how one type of gravimeter works. **2**
- (ii) The diagram shows features of two locations, *X* and *Y*, which affect gravity measurements taken at those locations. **3**



Justify the use of data reduction when conducting gravity surveys at *X* and *Y*.

- (e) Compare the information obtained about Earth's interior from natural vibrations and vibrations caused by human activity. In your answer, refer to the processes used to obtain the data. **6**

End of Question 31

Question 32 — Medical Physics (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- | | | | |
|-----|------|--|----------|
| (a) | (i) | In an endoscope, how do the arrangements of fibres in coherent and incoherent bundles differ? | 2 |
| | (ii) | Explain how light is used to create an endoscopic image. | 3 |
| (b) | (i) | Describe how ultrasound can be used to measure ONE property of bone. | 2 |
| | (ii) | B scans typically use ultrasound waves of frequency 20 MHz whereas sector scans typically use ultrasound waves of frequency 3.5 MHz. Account for the use of different frequencies in terms of the purposes of these scans. | 3 |
| (c) | | Explain how the application of a radio frequency wave changes the behaviour of nuclei with net spin in a strong, external magnetic field. | 4 |

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

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|-----|------|--|----------|
| (d) | (i) | Describe ONE way in which X-ray radiation for medical imaging is produced. | 2 |
| | (ii) | Why is a CAT scan image superior to a plain X-ray image? | 3 |
| (e) | | Functional imaging techniques allow assessment of the way the body is working, rather than providing detailed information about its structure. | 6 |

Relate the processes used to produce functional medical images to their benefits. Include examples in your answer.

Question 33 — Astrophysics (25 marks)

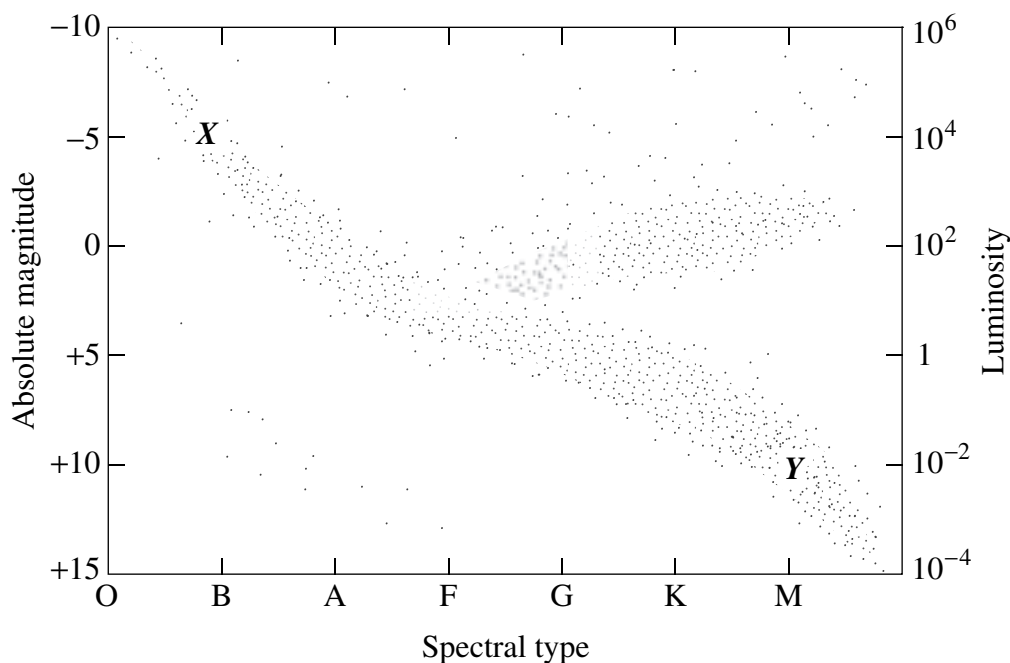
Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) Why do space-based optical telescopes provide more information than similar sized Earth-based optical telescopes? **2**

- (ii) Two stars, *P* and *Q*, have apparent magnitudes of 10.3 and 8.5. **3**

Use the brightness ratio to compare the brightness of these two stars.

- (b) (i) The diagram shows the positions of stars *X* and *Y* on a H–R diagram. **2**



Outline the differences in the spectra of stars *X* and *Y*.

- (ii) Describe a process which can be used to obtain the spectrum of an individual star. **3**

- (c) A star has a negative colour index. **4**

Explain how the colour index is determined and how it can be used to deduce information about this star.

Question 33 continues on page 34

Question 33 (continued)

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

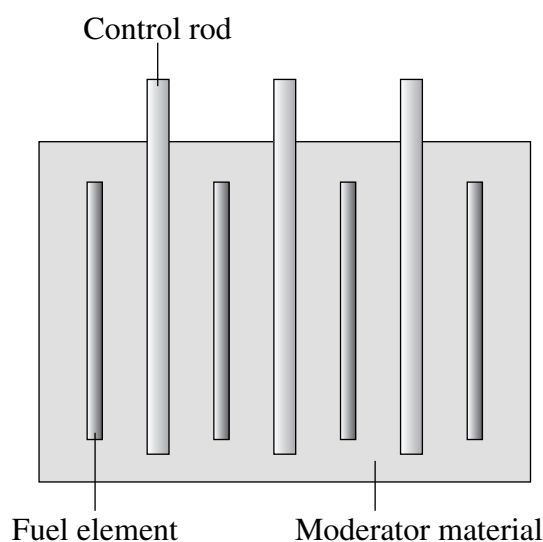
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|-----|------|---|----------|
| (d) | (i) | Use a labelled H–R diagram to show the areas in which stars would be plotted for a typical open cluster and a typical globular cluster. | 2 |
| | (ii) | Outline the differences between the stars in an open cluster and a globular cluster. | 3 |
| (e) | | Explain how gravitational forces affect star deaths for a range of solar masses. | 6 |

End of Question 33

Question 34 — From Quanta to Quarks (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) State the composition of the He-3 nucleus in terms of fundamental particles. **2**
- (ii) Outline features of the strong nuclear force. **3**
- (b) (i) Distinguish between the processes of nuclear fission and transmutation. **2**
- (ii) The diagram shows some components of a nuclear reactor. **3**



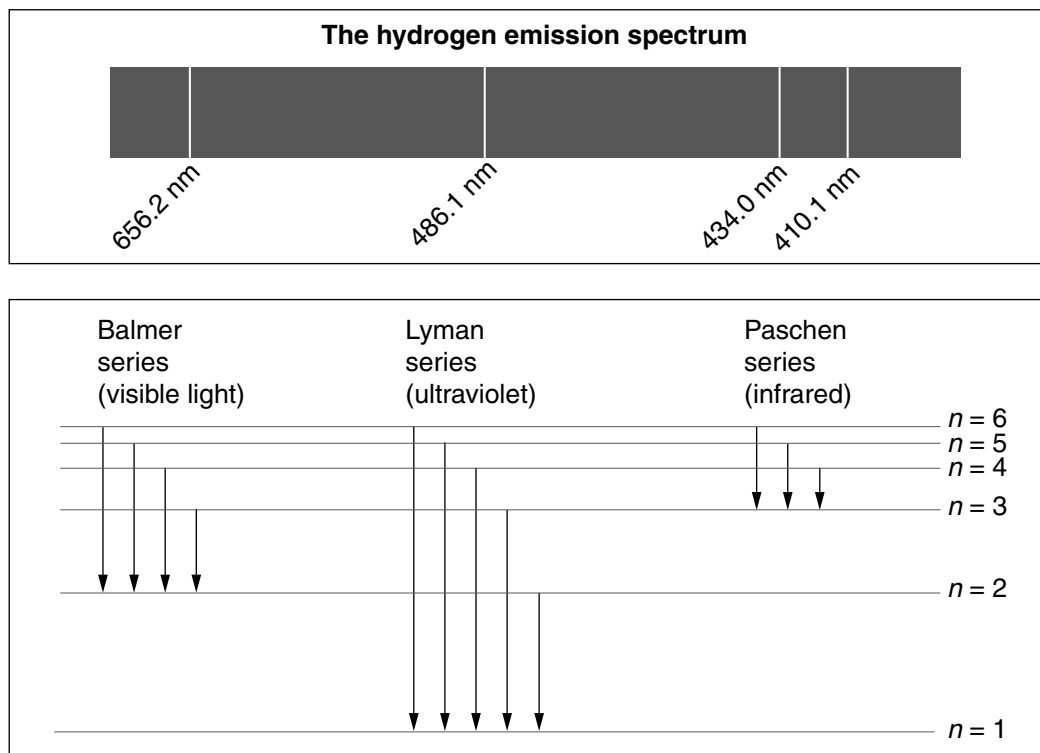
Explain how the labelled components work together to produce a controlled nuclear reaction.

Question 34 continues on page 36

Question 34 (continued)

- (c) The diagrams show features of the hydrogen emission spectrum.

4



With reference to Bohr's postulates, explain how the line at 434.0 nm in the hydrogen emission spectrum is produced. Support your answer with calculations.

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

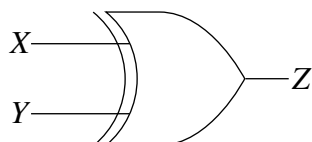
- (d) (i) How were the stable electron orbits in the Bohr model of the atom incompatible with a prediction of classical physics? 2
- (ii) How did de Broglie, and Davisson and Germer contribute to the modification of the Bohr model of the atom? 3
- (e) Explain how the properties of protons and neutrons contribute to their uses as tools for investigating matter. 6

End of Question 34

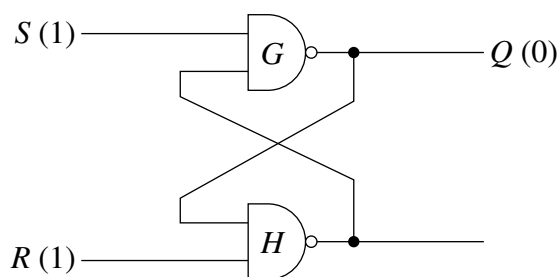
Question 35 — The Age of Silicon (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) The diagram shows a logic gate with two inputs. Construct a truth table for this logic gate. **2**



- (ii) The logic circuit shown can be used to store data. The Q output is in the '0' state. The R and S inputs are in the '1' state. **3**

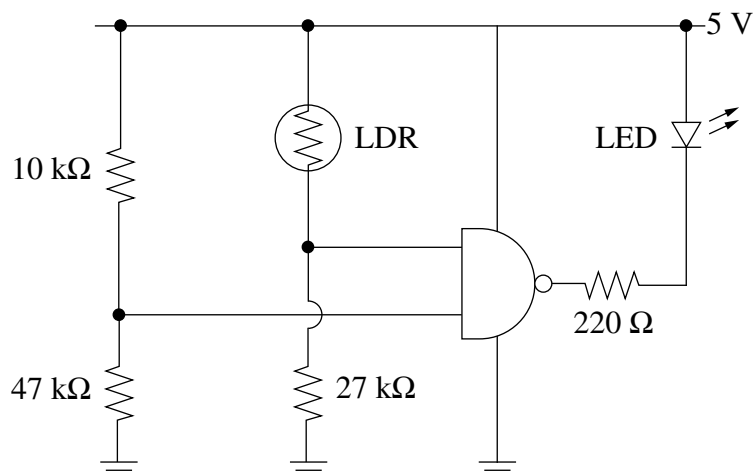


Show that the state of Q remains the same unless the S input state is changed to the '0' state.

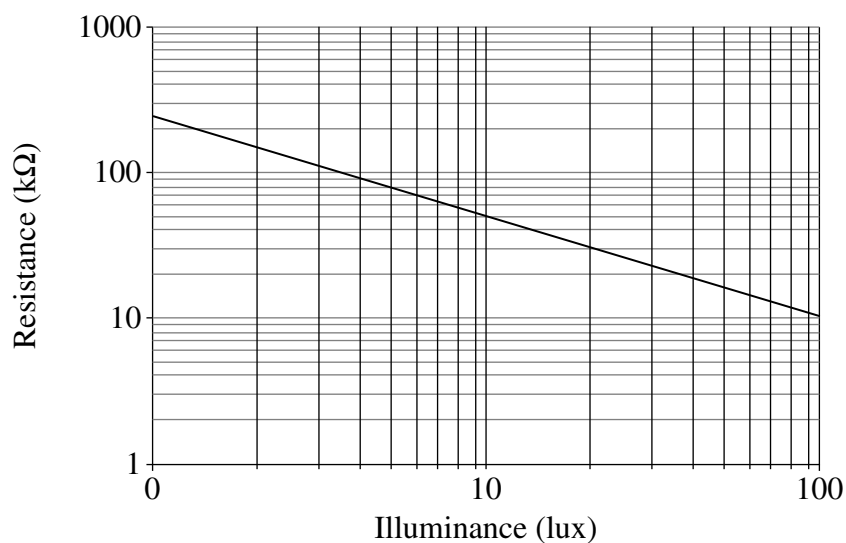
Question 35 continues on page 38

Question 35 (continued)

- (b) The diagram shows a circuit incorporating a light dependent resistor (LDR).



- (i) Use a sketch to show how the output of this circuit could be modified to operate a 12-volt fan. 2
- (ii) This graph shows how the resistance of the LDR changes with illuminance. 3



For an input logic level to be high, the voltage must be greater than 2 V.

What is the minimum illuminance required to turn on the LED? Support your answer with calculations.

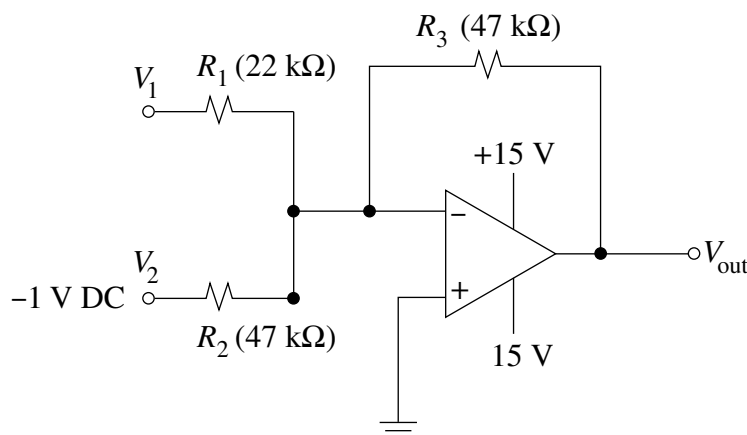
- (c) Contrast open-loop gain amplifiers and closed-loop gain amplifiers in terms of their function and circuit construction. 4

Question 35 continues on page 39

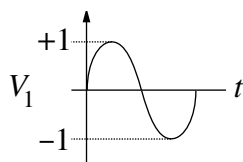
Question 35 (continued)

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) The diagram shows an operational amplifier connected to two input signals V_1 and V_2 .

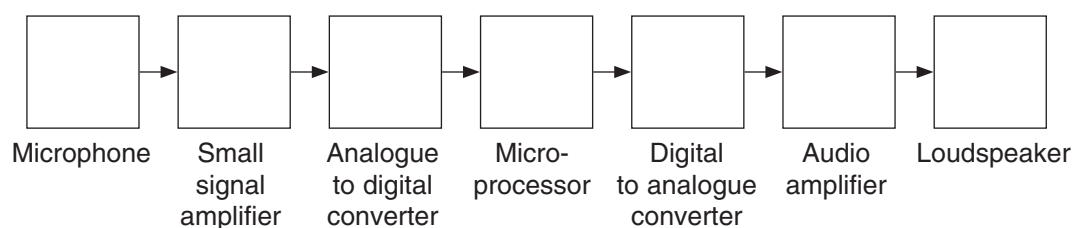


- (i) Determine the formula relating inputs V_1 and V_2 to V_{out} . 2
- (ii) The input to V_1 is shown by the graph. 3



Draw a voltage–time graph showing the voltage at V_{out} .

- (e) The block diagram shows the essential components of a digital signal processing system. 6



With reference to the diagram provided, explain why both analogue and digital circuits are used in audio systems.

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Physics

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \quad \text{therefore} \quad a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log_{10} \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium																							
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number Symbol		79 Au 197.0 Gold		Standard Atomic Weight Name		5 B 10.81 Boron		6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon																			
11 Na 22.99 Sodium		12 Mg 24.31 Magnesium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.38 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton									
37 Rb 85.47 Rubidium		38 Sr 87.61 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc 99.44 Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po 209.0 Polonium		85 At 210.0 Astatine		86 Rn 222.0 Radon	
87 Fr 223.0 Francium		88 Ra 226.0 Radium		89–103 Actinoids		104 Rf 261.1 Rutherfordium		105 Db 262.1 Dubnium		106 Sg 266.1 Seaborgium		107 Bh 264.1 Bohrium		108 Hs 277.1 Hassium		109 Mt 268.1 Meitnerium		110 Ds 271.1 Darmstadtium		111 Rg 272.1 Roentgenium		112 Cn 285.1 Copernicium		113 Nh 284.1 Nihonium		114 Fl 289.1 Flerovium		115 Mc 288.1 Moscovium		116 Lv 293.1 Livermorium		117 Ts 294.1 Tennessine		118 Og 294.1 Oganesson	

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm 144.9 Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac 227.0 Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np 237.0 Neptunium	94 Pu 244.0 Plutonium	95 Am 243.0 Americium	96 Cm 247.0 Curium	97 Bk 247.0 Berkelium	98 Cf 251.0 Californium	99 Es 252.0 Einsteinium	100 Fm 257.0 Fermium	101 Md 288.1 Mendelevium	102 No 289.1 Nobelium	103 Lr 260.1 Lawrencium
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Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

2017 HSC Physics Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	A
2	B
3	A
4	B
5	D
6	B
7	D
8	C
9	A
10	A
11	C
12	B
13	C
14	A
15	B
16	D
17	B
18	C
19	C
20	D

Section I, Part B

Question 21 (a)

Criteria	Marks
• Applies correct process to calculate the frequency	2
• Substitutes into a relevant formula	1

Sample answer:

$$c = f\lambda$$

$$f = \frac{3 \times 10^8}{550 \times 10^{-9}}$$

$$f = 5.45 \times 10^{14} \text{ Hz}$$

Question 21 (b)

Criteria	Marks
• Provides a valid explanation supported by relevant calculations	3
• Provides a valid explanation AND/OR • Shows some relevant calculation	2
• Provides some relevant information	1

Sample answer:

For an electron to be ejected the energy of the incoming photons must be greater than $5 \times 10^{-19} \text{ J}$.

Hence: $E_{\text{photon}} > 5 \times 10^{-19}$

$$\begin{aligned} E_{\text{photon}} &= hf \\ &= 6.62 \times 10^{-34} \times 5.46 \times 10^{14} \\ &= 3.63 \times 10^{-19} \text{ J} \end{aligned}$$

Therefore the electrons will not be ejected since $hf < 5 \times 10^{-19} \text{ J}$.

Question 22 (a)

Criteria	Marks
• Provides TWO ways of increasing torque	2
• Provides ONE way of increasing torque	1

Sample answer:

The torque can be increased by increasing the applied force or by increasing the distance between the nut and the point of the applied force.

Question 22 (b)

Criteria	Marks
• Applies correct process to calculate torque and indicates the correct direction	3
• Applies correct process to calculate torque OR • Shows some relevant working for calculating the torque and indicates the correct direction	2
• Substitutes into a relevant formula OR • Indicates the correct direction	1

Sample answer:

Side BC : $F = BIl \sin \theta = 0.2 \times 7 \times 0.08 \times \sin 90^\circ = 0.112 \text{ N}$

For 15 turns, the total force = $0.112 \text{ N} \times 15 = 1.68 \text{ N}$.

$\tau = Fd = \text{total force} \times \text{total distance}$, where total distance is the distance from the axis of rotation to point B (0.03 m)

Therefore $\tau = 1.68 \text{ N} \times 0.03 \text{ m} = 0.05 \text{ Nm}$ into the page.

Question 23

Criteria	Marks
• Provides an explanation of how theories in science are validated in different ways using examples relating to the theory of special relativity	5
• Provides an explanation of how theories in science are validated using at least one relevant example	4
• Provides relevant example(s) and/or explains how theories in science are validated	3
• Identifies a relevant example or outlines how theories in science are validated	2
• Provides some relevant information	1

Sample answer:

A theory must be consistent with observations, be supported by valid experiments and be useful in making predictions.

Time dilation has been observed in experiments using atomic clocks in aircraft in relation to synchronised clocks observed on the surface of the Earth. The results from such experiments can be used to predict the relativistic effects on sub-atomic particles travelling at high velocity in linear accelerators.

Question 24 (a)

Criteria	Marks
• Correctly calculates the escape velocity	2
• Substitutes into a relevant formula	1

Sample answer:

$$\begin{aligned}
 v &= \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 6.39 \times 10^{23}}{3.39 \times 10^6}} \\
 &= 5014.5 \text{ ms}^{-1} \\
 &= 5010 \text{ ms}^{-1} \text{ (rounded)}
 \end{aligned}$$

Question 24 (b)

Criteria	Marks
• Shows that the escape velocity is independent of mass using the law of conservation of energy	3
• Relates the conservation of energy to the escape velocity of an object	2
• Provides some relevant information	1

Sample answer:

Escape velocity is the minimum velocity required for an object to attain an E_p of 0.

Hence escape velocity is attained when $E_k + E_p = 0$.

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = 0$$

Rearranging:

$$mv^2 = \frac{2GMm}{r} \therefore \text{the escape velocity is independent of the mass of the object.}$$

Question 25 (a)

Criteria	Marks
• Outlines the contribution	2
• Provides some relevant information	1

Sample answer:

The Braggs aimed X-rays at materials and observed the diffraction patterns produced. This gave them information about arrangement and spacing of atoms, giving us clues about the crystal structure of materials.

Question 25 (b)

Criteria	Marks
• Explains how doping pure silicon with boron changes its electrical properties	2
• Provides some relevant information	1

Sample answer:

Boron from group 3 contributes a positive hole to the lattice of silicon. This increases the conductivity of silicon as holes are more easily able to move through the lattice.

Question 25 (c)

Criteria	Marks
• Outlines reasons for the use of superconductors on board maglev trains	3
• Outlines a reason OR • Identifies reasons	2
• Provides some relevant information	1

Sample answer:

Superconductors allow for the production of strong magnetic fields to cause the train to levitate above track magnets. The superconductors are light enough to be carried on board the train.

The refrigeration units required to produce the superconductive properties must be carried on the train, rather than cooling the whole rail track, in order to keep energy consumption within practical limits.

Question 26 (a)

Criteria	Marks
• Outlines consequences	2
• Outlines a consequence	1

Sample answer:

Upon entering Earth's atmosphere it will decelerate too rapidly for the astronauts to survive and the temperature reached will be dangerously high.

Question 26 (b)

Criteria	Marks
• Outlines the need to decrease E_p and/or E_k for a safe landing • Relates the information on the graph to the conversion of E_p and/or E_k	4
• Makes some relevant interpretation of the graph and/or recognizes the energy level desirable for a safe landing	2–3
• Provides some relevant information	1

Sample answer:

The graph shows that the gases around the shuttle, heated to 5000 K, radiate energy predominantly as light. The source of this energy is the conversion of the shuttle's potential and kinetic energy, reducing these so that the shuttle can make a safe landing.

Question 27 (a)

Criteria	Marks
• Applies correct process to calculate the maximum magnetic field strength	2
• Provides some relevant information	1

Sample answer:

B (flux density) is given by $\frac{\text{flux}}{\text{area}} \left(B = \frac{\Phi}{\pi r^2} \right)$

$$\begin{aligned} \text{Max } B &= \frac{0.6}{\pi \times (0.3)^2} \\ &= 2.1 \text{ T} \end{aligned}$$

Question 27 (b)

Criteria	Marks
Applies correct process to calculate the maximum voltage and identifies correct polarity of the terminals (P and Q)	3
- Applies correct process to calculate the maximum voltage OR - Shows some relevant calculations and identifies the polarity of the terminal(s)	2
Provides some relevant information	1

Sample answer:

Voltage (emf) = time rate of change of flux. From the graph, this is maximum when the steepest gradient occurs, ie $t = 10 - 12s$.

$$emf = \frac{-d\Phi}{dt} = -\left(\frac{-0.6}{2}\right) = 0.3V$$

Terminal P is positive and terminal Q is negative (consistent with Lenz's Law).

Question 28

Criteria	Marks
Contrasts the design of transformers and magnetic braking systems in terms of the effects that eddy currents have in each of these devices	6
Contrasts the design of transformers and magnetic braking systems with some reference to the effects that eddy currents have in these devices	5
- Outlines design features of transformers and magnetic braking systems - Identifies at least one effect that eddy currents have in one of these devices	4
Identifies features of transformers and/or magnetic braking systems and/or the effects of eddy currents	2–3
Provides some relevant information	1

Sample answer:

A transformer uses changes in magnetic flux to induce current in a secondary coil. An iron core increases the flux but iron, being a conductor, will experience the induction of unwanted eddy currents, causing resistive heating losses. Laminating the iron core minimises the current pathway, thus minimising eddy currents and subsequent heating losses.

In contrast, magnetic braking systems need to maximise eddy currents since magnetic braking relies on converting kinetic energy to electrical current, and subsequently to heat. In order to maximise the current, magnetic braking systems need to provide substantial current pathway. This is done by providing large sheets or bulk volumes of conductive material, such as copper.

Question 29 (a)

Criteria	Marks
• Applies correct process to calculate the launch velocity with appropriate units	4
• Applies conservation of energy law to calculate the launch velocity with a minor error	3
• Shows some relevant calculation steps	2
• Substitutes into a relevant formula	1

Sample answer:

$$k = \frac{24}{0.08} = 300$$

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

$$\frac{1}{2} \times 0.04 \times v^2 = \frac{1}{2} \times 300 \times 0.08^2$$

$$v^2 = 300 \times 0.08 \times 2$$

$$v = 6.9 \text{ ms}^{-1}$$

Answers could include:

Calculating the area under the curve for work done.

Question 29 (b)

Criteria	Marks
• Applies correct process to calculate the range	3
• Shows some relevant calculation steps	2
• Provides some relevant information	1

Sample answer:

The range is equal to the horizontal velocity component $\times$ the time of flight.

The time of flight is twice the time taken to reach the highest point.

Since $v = u + at$, at the highest point $v = 0 = 10 \sin 60^\circ - 9.8 t$

Hence time of flight $t = \frac{20 \sin 60^\circ}{9.8} = 1.77 \text{ s}$.

Hence the range $= 10 \times \cos 60^\circ \times 1.77 = 8.8 \text{ m}$.

Question 30

Criteria	Marks
• Provides an analysis of the motion of the proton	4
• Describes the path and/or direction of the proton	3
• Provides some characteristics of the path and/or direction of the proton	2
• Provides some relevant information	1

Sample answer:

The magnetic field will cause the proton to move in a circular path, initially moving out of the page, continuing in an anti-clockwise direction as viewed from the right. At the same time, the electric field will cause the proton to move to the left with increasing speed. The resultant motion of the proton is the vector sum of these two components, which will look like a helix that is getting stretched out towards the left. This helix will be decreasing in radius as the proton loses energy as it radiates electromagnetic radiation as it is an accelerating charge.

[Answers may include a labelled diagram.]

Section II

Question 31 (a) (i)

Criteria	Marks
• Gives TWO reasons	2
• Gives ONE reason	1

Sample answer:

Wegener's notion of drifting continents went against existing dogma and so there was reluctance to accept his new notions.

Wegener's explanations for continental drift were inadequate in accounting for the scale of the movements of the continents and alternative explanations could be provided for observations so there was no pressure to change.

Question 31 (a) (ii)

Criteria	Marks
• Relates features associated with the magnetic properties of the oceanic crust to the theory of plate tectonics	3
• Relates magnetism to plate tectonics and/or mid-ocean ridges and/or polar wandering	2
• Provides some relevant information	1

Sample answer:

Analysis of magnetic reversals in the crustal rocks showed a symmetry on either side of mid-ocean ridges that was consistent with seafloor spreading.

The increasing age of rocks associated with increasing distance from the mid-ocean ridges was consistent with seafloor spreading and plate tectonics.

Answers could include:

Relationship between magnetism and polar wandering.

Question 31 (b) (i)

Criteria	Marks
• Contrasts the movement of particles associated with the waves	2
• Identifies the movement of particles associated with one of the waves	1

Sample answer:

The particles transferring energy in the form of a *P* wave oscillate parallel to the direction of energy propagation whereas the particles transferring energy in the form of an *S* wave oscillate perpendicular to the direction of energy propagation.

Question 31 (b) (ii)

Criteria	Marks
• Describes the behaviour of the <i>P</i> waves and <i>S</i> waves	3
• Provides characteristics of the behaviour of the <i>P</i> waves and/or <i>S</i> waves	2
• Provides some relevant information	1

Sample answer:

At the boundary both *P* and *S* waves will be reflected at the same angle as the angle of incidence.

Only *P* waves will cross the boundary and travel into the liquid.

P waves will be refracted as they cross the solid–liquid boundary.

Question 31 (c)

Criteria	Marks
- Describes two methods - Refers to the relevant equations	4
Outlines two methods	3
Outlines a method or identifies two methods	2
Provides some relevant information	1

Sample answer:

One method to calculate the mass of Earth is to use the relationship between the strength of the gravitational field and the radius of Earth and its mass. Gravitational field strength and the radius of Earth can be experimentally determined.

$$F = mg = \frac{GMm}{r^2}$$

$$g = \frac{GM}{r^2}.$$

Hence

$$M = \frac{gr^2}{G}.$$

Another method is to apply Kepler's law using data – radius and period – from a satellite's orbit. Both the radius and period of a satellite can be experimentally determined.

$$\frac{r^3}{T^2} = \frac{GM}{r\pi^2}.$$

Hence

$$M = r\pi^2 \times \frac{r^3}{GT^2}.$$

Answers could include:

Pendulum.

Question 31 (d) (i)

Criteria	Marks
• Outlines how one type of gravimeter works	2
• Provides some relevant information	1

Sample answer:

One type of gravimeter uses the inertial properties of a mass confined by springs in a vacuum. The effect of gravity on the confined mass causes it to move independently of the surrounding housing. This movement is calibrated against variations in the strength of gravity so that the instrument can be used to directly measure the strength of gravity.

Question 31 (d) (ii)

Criteria	Marks
• Justifies why data reduction is necessary in the given scenario	3
• Shows some understanding of the effect of location on gravimeter measurements	2
• Provides some relevant information	1

Sample answer:

Data reduction removes the effects on the measurement of gravity which are caused by features associated with specific locations. This allows valid comparisons between these locations, independent of these features.

The effect on gravimeter readings of the rotation of Earth on its axis is greater at the equator (X) than at higher latitudes (Y), reducing the readings more at the equator.

At higher altitudes (Y) the reduction on the gravimeter readings is greater than at lower altitudes (X).

Question 31 (e)

Criteria	Marks
- Shows the similarities and/or differences between the information obtained from natural vibrations and vibrations caused by human activity - Refers to the processes used to obtain the data	6
- Outlines the information obtained from natural vibrations and vibrations caused by human activity - Refers to the process(es) used to obtain the data	4–5
Shows some understanding of obtaining information from natural vibrations and/or vibrations caused by human activity	2–3
Provides some relevant information	1

Sample answer:

The positions, sizes, shapes and types of material associated with structures into the Earth can be deduced from vibrations caused by human activity. These can be produced in different ways at or near Earth's surface using geophones or seismographs. The energy is reflected from relatively close to Earth's surface by layers of different materials.

Natural vibrations originate from seismic activity including earthquakes and volcanoes. The energy from these is generally much greater than vibrations caused by human activity and so can be detected by seismographs located at hundreds of locations on Earth's surface. Analysis of these, as well as knowledge about the behaviour of different types of waves, provides information about Earth's interior, including its layered structure, the presence of solids and liquids and variations in density.

Question 32 (a) (i)

Criteria	Marks
Shows an understanding of how the arrangements differ	2
Provides some relevant information	1

Sample answer:

In coherent bundles the position of individual fibres relative to each other is the same at each end. In incoherent bundles relative position is not maintained.

Question 32 (a) (ii)

Criteria	Marks
• Explains how light is used to create an endoscopic image	3
• Shows some understanding of endoscopy	2
• Provides some relevant information	1

Sample answer:

Endoscopy relies on the physical principle of total internal reflection. Light is reflected within each fibre and thereby manipulated. The incoherent bundles provide illumination to the site, which would otherwise be in darkness. This then permits the coherent bundles to transmit an image to a camera for external viewing.

Question 32 (b) (i)

Criteria	Marks
• Describes how ultrasound can be used to measure ONE property of bone	2
• Provides some relevant information	1

Sample answer:

Measurement of bone density can be made by directing ultrasound waves at a bone (eg heel bone) and measuring the amount of waves transmitted. The higher the bone density, the more ultrasound waves are absorbed and therefore fewer transmitted.

Question 32 (b) (ii)

Criteria	Marks
• Accounts for the different frequencies of the scans in terms of their purposes	3
• Outlines the purposes of the scans OR • Outlines the purpose of one scan and relates its purpose to the choice of frequency	2
• Provides some relevant information	1

Sample answer:

B scans are used to examine eyes and therefore have no need to penetrate tissue more than a few centimetres. Therefore a higher frequency can be used to obtain greater detail. Sector scans are used to examine tissues deep within the body eg pregnant uterus. By using a lower frequency, a greater depth of tissue can be examined, albeit with less detail.

Question 32 (c)

Criteria	Marks
• Explains how the application of a radio frequency wave changes the behaviour of nuclei in B_0	4
• Outlines behaviour of nuclei in B_0 and/or effects of radio frequency wave	2–3
• Provides some relevant information	1

Sample answer:

Nuclei with net spin line up with the external magnetic field, B_0 . Their orientation may be in one of two ways – lower energy spin up or higher energy spin down – with slightly more in the lower energy state. At the same time, these nuclei precess around the direction on B_0 with a frequency (Larmor frequency) that is dependent on the magnitude of B_0 , the type of nucleus and its immediate environment. When a radio frequency corresponding to the Larmor frequency is applied, the lower energy spin up nuclei absorb the energy and flip to spin down orientation, then precess in phase with each other. Nuclei which precess at different frequencies are unaffected.

Question 32 (d) (i)

Criteria	Marks
• Describes production of X-rays by one method	2
• Provides some relevant information	1

Sample answer:

Electrons are accelerated to high velocities by a high voltage. When they strike the tungsten target, their rapid deceleration produces EMR in the X-ray range of the spectrum.

Question 32 (d) (ii)

Criteria	Marks
• Shows an understanding of why a CAT scan image is superior to an X-ray image	3
• Identifies features of CAT scan image and/or X-ray image	2
• Provides some relevant information	1

Sample answer:

X-ray images are formed by the shadows cast on a film (or digital detector) in which each point on the image represents all of the tissue in front it, often several centimetres in depth. CAT scans use tomography to gather information on X-ray attenuation that can be organised into ‘slices’. This provides a clearer image as there is no superimposition of other layers.

Question 32 (e)

Criteria	Marks
- Relates the processes that produce functional medical images to benefits - Includes relevant examples	6
- Outlines how functional image(s) are produced - Outlines benefit(s) of functional imaging - Includes example(s)	4–5
- Outlines benefit(s) of functional imaging AND/OR - Outlines how functional image(s) are produced AND/OR - Includes example(s)	2–3
Provides some relevant information	1

Sample answer:

PET scans and bone scans using gamma radiation are two functional imaging techniques. In both of these the uptake of radioisotopes used to produce the images is dependent on how the body is working.

In PET for example, the uptake of an injected radioactive isotope (FDG) is greater in, for example a brain tumour, than in healthy tissue and so tumours can be detected on the images produced from the radiation emitted from the concentrated radioisotope.

In a bone scan, molecules tagged with gamma-emitting Tc-99m accumulate in tumours in the bone and this is beneficial as the abnormal tissue can be detected clearly on the bone scan images.

Functional imaging techniques can make abnormal processes that cannot be detected using structural scans visible, thus allowing for earlier detection and more effective treatment.

Answers could include:

Doppler ultrasound or fMRI.

Question 33 (a) (i)

Criteria	Marks
• Shows an understanding of why space-based optical telescopes provide more information	2
• Provides some relevant information	1

Sample answer:

Space-based telescopes reduce both atmospheric absorption of specific frequencies (electromagnetic radiation) and atmospheric distortion of the light from stars due to turbulence.

Question 33 (a) (ii)

Criteria	Marks
• Correctly uses the brightness ratio to make a comparison	3
• Correctly applies the brightness ratio	2
• Provides some relevant information	1

Sample answer:

$$\begin{aligned}\frac{I_Q}{I_P} &= 100^{\frac{(m_P - m_Q)}{5}} \\ &= 100^{\frac{(10.3 - 8.5)}{5}} \\ &= 100^{0.36} \\ &= 5.25\end{aligned}$$

Star *Q* is 5.25 times brighter than star *P*.

Question 33 (b) (i)

Criteria	Marks
• Outlines the differences	2
• Provides some relevant information	1

Sample answer:

Spectrum for Star *X* shows highest intensity in the blue part of the spectrum (short wavelength) with ionised He lines and strong UV component.

Spectrum for Star *Y* shows highest intensity in the red part of the spectrum (long wavelength) with dominant molecular lines and strong neutral metal lines.

Question 33 (b) (ii)

Criteria	Marks
• Describes a suitable process	3
• Outlines some features of a suitable process	2
• Provides some relevant information	1

Sample answer:

A diffraction grating is placed over the opening of a telescope. The resulting photographic image shows multiple spectra, one for each identified star in the field of view.

Answers could include:

Optic fibres are attached to a platen specifically where starlight falls for a particular field of view. The starlight is then passed to a spectroscope via the optic fibre for individual stars.

Question 33 (c)

Criteria	Marks
• Explains how the colour index is determined	4
• Explains how the colour index can be used to deduce information about the star	
• Shows some understanding of the colour index and/or the characteristics of the star	2–3
• Provides some relevant information	1

Sample answer:

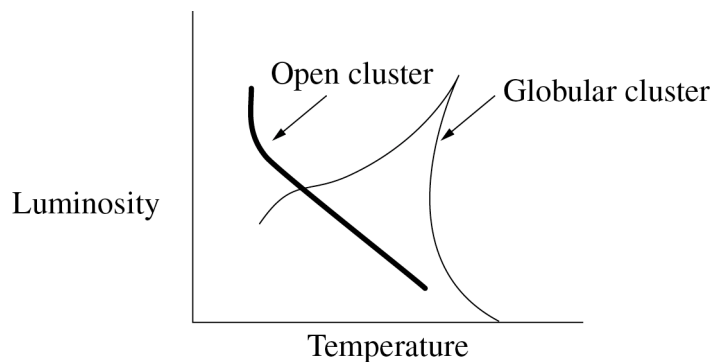
Colour index is calculated using $CI = B - V^*$ where B is the apparent magnitude measured when the star's light is passed through a blue filter and V is the visual apparent magnitude measured using a greenish-yellow filter (peak visual λ).

The more negative the CI, the hotter and hence more blue the star. Stars having $CI < 0$ have surface temperatures $> 10\,000\text{ K}$.

Question 33 (d) (i)

Criteria	Marks
• Shows both clusters on a H–R diagram	2
• Shows one cluster on a H–R diagram	1

Sample answer:



Question 33 (d) (ii)

Criteria	Marks
• Outlines differences between stars from both cluster types	3
• Identifies features of the stars in an open cluster and/or a globular cluster	2
• Provides some relevant information	1

Sample answer:

Open clusters consist of relatively young, main-sequence stars where as globular clusters contain older stars at a post-main-sequence evolutionary stage.

Opens clusters generally contain stars that are hotter than those in globular clusters.

Question 33 (e)

Criteria	Marks
Explains how gravitational forces affect star deaths for a range of solar masses	6
Outlines the effects of gravitational forces on star deaths for a range of solar masses	4–5
- Recognises the relationship between a star's death and its mass AND/OR - Identifies the effects of gravitational forces on star deaths	2–3
Provides some relevant information	1

Sample answer:

Gravity causes stars having a mass of less than 1.4 solar masses to collapse to form a white dwarf when nuclear fusion ceases.

Stars having initial masses of from 1.4 to 8 solar masses expel some of their mass into space during their evolution to star death, ultimately leaving a type of star dependent on the mass and hence gravitational forces.

For stars having a final mass of 1.4 to 3 solar masses, once fusion ceases, gravity causes the protons and electrons to form neutrons and a neutron star/pulsar is formed.

For stars having final masses greater than 3 solar masses, gravitational forces are stronger causing the star to ultimately collapse and form a black hole.

Question 34 (a) (i)

Criteria	Marks
Identifies type and number of nucleons and number of quarks	2
Provides some relevant information	1

Sample answer:

Helium 3 nucleus consists of:

- 1 × neutron
- 2 × protons

which themselves are made up of three quarks.

Question 34 (a) (ii)

Criteria	Marks
• Outlines features of the strong nuclear force	3
• Identifies features of the strong nuclear force or outlines a feature	2
• Identifies a feature of fundamental forces	1

Sample answer:

The strong nuclear force is much stronger than the coulomb repulsion and acts over only very small distances.

The strong nuclear force is extremely attractive between nucleons about 1fm apart but decreases rapidly at distances greater than about 2.5fm. This force becomes repulsive at distances less than 0.7fm.

Question 34 (b) (i)

Criteria	Marks
• Distinguishes between fission and transmutation	2
• Identifies a feature of fission or transmutation	1

Sample answer:

A transmutation involves an atomic nucleus turning into that of another element by radioactive decay or nuclear bombardment. Fission involves the splitting of an atom into smaller, similar sized fragments due to the capturing of a neutron. This process releases energy.

Question 34 (b) (ii)

Criteria	Marks
• Explains how the labelled components are used to produce a controlled nuclear reaction	3
• Outlines how some of the labelled components are used to produce a controlled nuclear reaction	2
• Identifies the role of a labelled component	1

Sample answer:

Fuel elements are a fissile material that releases energy and high-speed neutrons when atoms are split. The moderator slows the neutrons down so that they can be captured by more fissile atoms, which then split. The control rods absorb neutrons to limit the number of atoms undergoing fission, thereby controlling the release of energy.

Question 34 (c)

Criteria	Marks
- Explains the production of the line with reference to Bohr's postulates - Supports answer with calculations	4
- Relates the production of the line to Bohr's postulates - Includes some relevant calculations	3
- Shows some understanding of Bohr's postulates AND/OR - Provides some relevant calculations	2
Provides some relevant information	1

Sample answer:

Bohr's postulates state that an electron orbits in fixed energy levels. Only when an electron jumps from a higher energy level to a lower energy level, it emits an exact amount of energy in the form of a photon. The specific wavelength of the photon can be determined according to the Rydberg equation. The blue light is part of the visible Balmer series and therefore $n_f = 2$.

The wavelength in this case is 434.0 nm. According to the equation, this corresponds to a jump from the 5th energy level to the 2nd energy level.

For Balmer series $n_f = 2$

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right)$$

$$n_i^2 = \frac{1}{\left(\frac{1}{4} - \frac{1}{1.097 \times 10^7 \times 434 \times 10^9} \right)}$$

$$n_i = 5$$

Answers could include:

Taking clues from the diagram of the Balmer series, student could put $n_f = 2$ and $n_i = 5$ into the equation to obtain $\lambda = 434.0$ nm.

Question 34 (d) (i)

Criteria	Marks
Accounts for the failure of classical physics in explaining stable electron orbits in the Bohr model of the atom	2
Provides some relevant information	1

Sample answer:

The stable electron orbits necessary for the Bohr model to work assumed that the electron would be travelling in circular motion about the nucleus. This would mean that the electron is constantly accelerating. But since accelerating charges give out electromagnetic radiation, the electron would lose energy and spiral in towards the nucleus.

Question 34 (d) (ii)

Criteria	Marks
- Shows an understanding of how de Broglie's model explained the stable electron orbits - Outlines Davisson and Germer's role in validating de Broglie's model	3
Shows an understanding of de Broglie's model AND/OR Davisson and Germer's role in validating this model	2
Provides some relevant information	1

Sample answer:

Since classical physics could not explain the stable electron orbits in the Bohr model, de Broglie modelled the electrons existing as standing waves about the nucleus with each of Bohr's stationary states (energy levels) being associated with various wavelengths. Hence only integer multiples of the fundamental wavelength were possible.

This idea was then validated by Davisson and Germer through their electron scattering experiment which provided evidence that electrons were waves.

Question 34 (e)

Criteria	Marks
• Outlines properties of protons and neutrons • Outlines the uses of protons and neutrons as tools for investigating matter • Relates the properties to the uses of protons and neutrons as tools for investigating matter	6
• Outlines properties of protons and/or neutrons • Outlines the uses of protons and/or neutrons as tools for investigating matter • Provides some link between the properties	4–5
• Identifies some properties of protons and/or neutrons AND/OR • Identifies uses of protons and/or neutrons as tools for investigating matter	2–3
• Provides some relevant information	1

Sample answer:

Protons and neutrons have almost identical masses but the proton is positively charged and the neutron has no charge.

Having a positive charge allows protons to be accelerated to very high velocities in particle accelerators. The protons then collide head-on with other protons, transferring their energy to mass ($E = mc^2$) and producing other particles which can be detected through their interaction with large detectors. This allows us to discover new particles and study the fundamental nature of matter.

In studying normal matter, protons cannot get close enough since they are repelled by the positive nucleus. Instead, neutron beams can be fired at nuclei in a process known as neutron scattering. Since all matter has a dual particle/wave nature, neutrons, having a very small de Broglie wavelength can produce diffraction patterns giving us detailed models of atomic and molecular structures.

Question 35 (a) (i)

Criteria	Marks
• Draws the correct truth table	2
• Provides some relevant information	1

Sample answer:

Truth table shown for exclusive OR gate.

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

Question 35 (a) (ii)

Criteria	Marks
• Shows that the state of Q only changes when S is changed to a 0	3
• Shows some understanding of the operation of the circuit	2
• Provides some relevant information	1

Sample answer:

The Q state is input to gate H. The output of gate H is always 1 regardless of the state of R.

Q	R	Output from H
0	1	1
0	0	1

The output of gate H is fed into gate G.

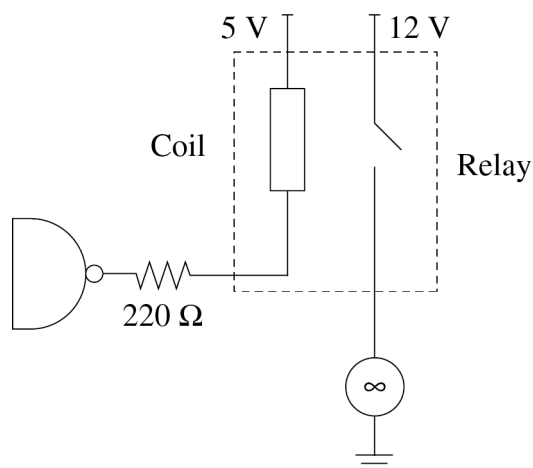
Output from H	S	Q (output from G)	
1	0	1	(Q changes from 0 to 1)
1	1	0	(no change to state)

So the state of Q will only change if the S input is changed to 0.

Question 35 (b) (i)

Criteria	Marks
• Draws a sketch showing a correct modification of the circuit	2
• Provides some relevant information	1

Sample answer:



Question 35 (b) (ii)

Criteria	Marks
• Determines the correct minimum illuminance	3
• Calculates the required resistance of the LDR	2
• Shows a basic understanding of the circuit OR • Substitutes into a relevant formula	1

Sample answer:

To switch the gate OFF so that it acts as a current sink the voltage across the 27 kΩ resistor must be greater than 2 V.

$$2\text{ V} = 5\text{ V} \times \frac{27\text{ k}\Omega}{R_{\text{LDR}} + 27\text{ k}\Omega}$$

$R_{\text{LDR}} = 40.5\text{ k}\Omega$ and from the graph the illumination threshold at which this occurs is about 12 lux.

Question 35 (c)

Criteria	Marks
• Contrasts the function and circuit construction of open-loop gain and closed-loop gain amplifiers	4
• Outlines the functions and/or circuit construction of open-loop gain and/or closed-loop gain amplifiers	2–3
• Provides some relevant information	1

Sample answer:

Open-loop gain refers to an operational amplifier when no feedback is provided. Typically, the open-loop gain is of the order of 200×10^3 . Any difference in voltage between the inverting and non-inverting input will cause the output to swing to the supply rail voltage. Using an amplifier in this mode is useful if the amplifier is used as a switch or in a circuit that is comparing two voltages and a binary output is required. However, where the shape of the waveform is important then a closed-loop amplifier is required. This means that a portion of the output voltage is fed back to the input as negative feedback. The effect of negative feedback is to reduce the gain of the amplifier.

Question 35 (d) (i)

Criteria	Marks
• Correctly shows the mathematical relationship between V_{in} and V_{out} and identifies the circuit as a summing amplifier	2
• Provides some relevant information	1

Sample answer:

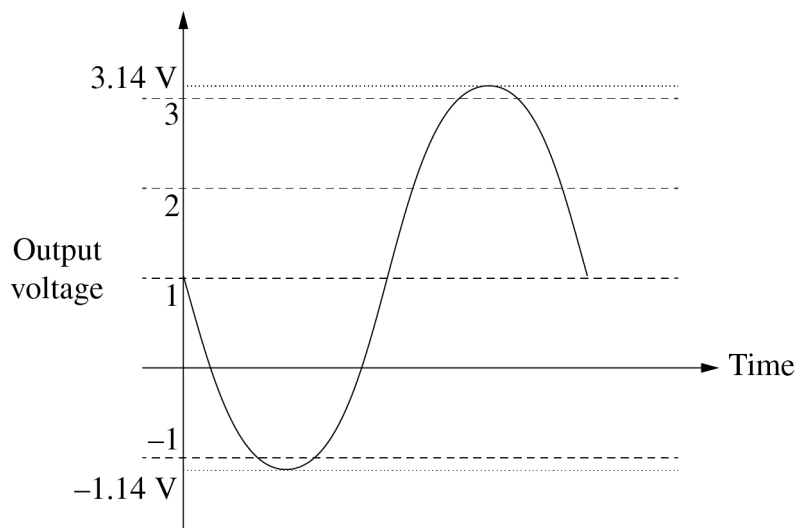
The circuit describes a summing inverting amplifier with the mathematical relationship between V_{in} and V_{out} equal to:

$$V_{out} = -R_3 \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right).$$

Question 35 (d) (ii)

Criteria	Marks
• Draws a correct graph with all relevant voltages and signal inversion	3
• Provides a partially correct graph	2
• Provides some relevant information	1

Sample answer:



Question 35 (e)

Criteria	Marks
• Explains why both analogue and digital circuits are used in audio systems • Makes reference to the diagram	6
• Outlines why both analogue and digital circuits are used in audio systems • Refers to the diagram	5
• Outlines why both analogue and digital circuits are used in audio systems	4
• Shows some understanding of the use of analogue and/or digital circuits in audio systems OR • Outlines some components of the system shown	2–3
• Provides some relevant information	1

Sample answer:

Audio systems must involve analogue signals. The value of using analogue circuit components, such as microphones and loudspeakers, is that the input and outputs must be analogue for humans to hear it. Although analogue circuits can perform signal processing including recording, amplification and reproduction, the value of converting the signal to a digital form (analogue–digital converter) is that in digital form it can be stored, reproduced and transmitted without loss of information. Developments in integrated circuits technology (miniaturisation, increased functionality) have allowed audio systems to be incorporated into a wide variety of communication systems such as TV, phones and internet.

2017 HSC Physics Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.3.3.3.3	H3
2	1	9.2.4.2.4	H6
3	1	9.4.1.2.4	H9
4	1	9.2.1.3.3	H6
5	1	9.4.3.2.3	H3
6	1	9.4.4.2.3	H1
7	1	9.2.4.2.2	H1
8	1	9.4.1.2.2	H9
9	1	9.4.1.2.2	H1
10	1	9.3.3.2.3	H3
11	1	9.3.4.3.2/9.4.1.3.3	H7
12	1	9.2.1.2.3	H7
13	1	9.3.1.2.5	H9
14	1	9.3.2.2.3	H7
15	1	9.2.2.3.4	H6
16	1	9.3.1.2.4	H9
17	1	9.3.1.3.3	H9
18	1	9.2.2.3.4/9.4.1.3.3	H9
19	1	9.3.1.2.1	H9
20	1	9.2.4.3.5	H6

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.4.2.3.4, 9.4.2.2.6	H8
21 (b)	3	9.4.2.3.4, 9.4.2.2.6, 9.4.2.2.5	H7
22 (a)	2	9.3.1.2.3	H6
22 (b)	3	9.3.1.2.3, 9.3.1.2.4, 9.3.1.2.5, 9.3.1.3.3	H9
23	5	9.2.4.3.4	H2
24 (a)	2	9.2.2.2.3	H9
24 (b)	3	9.2.2.2.3, 9.2.1.2.2, 9.2.1.2.3	H7
25 (a)	2	9.4.4.2.1, 9.4.4.1.1	H1
25 (b)	2	9.4.3.2.6, 9.4.3.2.7	H3
25 (c)	3	9.4.4.3.4	H3
26 (a)	2	9.2.2.2.2.13	H6, H7
26 (b)	4	9.2.2.2.12, 9.4.2.1.1	H7

Question	Marks	Content	Syllabus outcomes
27 (a)	2	9.3.2.2.2, 9.3.2.2.3	H7, H9
27 (b)	3	9.3.2.2.4	H9
28	6	9.3.2.3.4, 9.3.4.3.3	H7
29 (a)	4	9.2.1.2.2, 9.2.2.3.2	H7, H13
29 (b)	3	9.2.2.3.1, 9.4.1.2.5, 9.4.1.3.3	H7, H9
30	4	9.4.1.2.5, 9.4.1.3.3, 9.4.1.2.3	H9

Section II

Question	Marks	Content	Syllabus outcomes
Question 31		Geophysics	
(a) (i)	2	9.5.4.2.4	H1
(a) (ii)	3	9.5.4.2.2/3, 9.5.4.3.2	H1
(b) (i)	2	9.5.3.2.2	H7
(b) (ii)	3	9.5.3.2.2/3, 9.5.3.3.1	H8
(c)	4	9.5.2.3.3/4	H14
(d) (i)	2	9.5.2.2.6	H7
(d) (ii)	3	9.5.2.2.7/8, 9.5.2.3.5	H14
(e)	6	9.5.3.2.5/6/7, 9.5.3.3.2/3	H14
Question 32		Medical Physics	
(a) (i)	2	9.6.2.2.6	H8
(a) (ii)	3	9.6.2.2.5/6	H8
(b) (i)	2	9.6.1.3.4	H7
(b) (ii)	3	9.6.1.2.7	H7
(c)	4	9.6.4.2.1/3/4/5/6	H9
(d) (i)	2	9.6.2.2.1	H8
(d) (ii)	3	9.6.2.2.3, 9.6.2.2.4, 9.6.2.3.2	H8
(e)	6	9.6.3.2.2/3/4/5	H4, H7, H8

Question 33		Astrophysics	
(a) (i)	2	9.7.1.2.4	H4
(a) (ii)	3	9.7.4.3.1	H12
(b) (i)	2	9.7.6.3.2	H14
(b) (ii)	3	9.7.3.2.2	H3
(c)	4	9.7.4.2.4	H3
(d) (i)	2	9.7.6.3.1	H14
(d) (ii)	3	9.7.6.3.2	H14
(e)	6	9.7.6.2.6	H13
Question 34		From Quanta to Quarks	
(a) (i)	2	9.8.4.2.5, 9.8.3.2.1	H10
(a) (ii)	3	9.8.3.2.8, 9.8.4.2.5	H10
(b) (i)	2	9.8.3.2.3, 9.8.3.3.2	H10
(b) (ii)	3	9.8.3.2.11	H3
(c)	4	9.8.1.2.5, 9.8.1.3.2, 9.8.1.3.3	H7, H10
(d) (i)	2	9.8.2.2.4	H1
(d) (ii)	3	9.8.2.2.3, 9.8.2.2.4, 9.8.2.2.1, 9.8.4.2.4	H2
(e)	6	9.8.4.2.3	H3
Question 35		The Age of Silicon	
(a) (i)	2	9.9.5.2.1, 9.9.5.3.1	H3, H13
(a) (ii)	3	9.9.5.3.2	H3, H13
(b) (i)	2	9.9.4.2.1, 9.9.4.3.1, 9.9.4.3.2	H7, H11, H14
(b) (ii)	3	9.9.2.2.4, 9.9.2.2.5, 9.9.2.3.3	H4, H12
(c)	4	9.9.6.2.1, 9.9.6.2.5, 9.9.6.2.9	H13, H14
(d) (i)	2	9.9.6.2.8, 9.9.6.2.11, 9.9.6.3.3	H12, H13, H14
(d) (ii)	3	9.9.6.2.8, 9.9.6.3.2, 9.9.7.3.4	H11, H12, H13
(e)	6	9.9.1.2.1, 9.9.1.2.4, 9.9.2.2.3	H1, H3, H7